Reduction of Order

- **Problem 1.** Find the GS of $y'' - 2ry' + r^2y = 0$ if one solution is $y_1 = e^{rx}$. For the 2nd solution, we will use reduction of order.
- Let $y_2(x) = v(x)y_1(X)$
- Here $P(x) = -2r$, so $e^{-\int P\,dx} = e^{2rx}$.

$$v(x) = \int \frac{e^{2rx}}{(e^{rx})^2}\,dx = \int 1\,dx = x$$

$$\boxed{y_2 = xe^{rx}}$$

- The GS is $y(x) = c_1e^{rx} + C_2xe^{rx}$.



Reduction of Order

- **Problem 2.** Given $y_1 = x$ solves $x^2y'' - xy' + y = 0$, $x > 0$, find y_2 and write the general solution.
- Standard form: $y'' - \frac{1}{x}y' + \frac{1}{x^2}y = 0 \Rightarrow P(x) = -\frac{1}{x}$
- $e^{-\int P\,dx} = e^{\int \frac{1}{x}\,dx} = x$

  $$v(x) = \int \frac{x}{x^2}\,dx = \int \frac{1}{x}\,dx = \ln x$$

  $$\boxed{y_2 = x \ln x}$$
- General solution: $y = c_1 x + c_2 x \ln x$.



Reduction of Order

- **Problem 3.** Given $y_1 = x$ solves $(1-x^2)y'' - 2xy' + 2y = 0$, find y_2.
- Standard form:
$$y'' - \frac{2x}{1-x^2}y' + \frac{2}{1-x^2}y = 0 \;\Rightarrow\; P(x) = -\frac{2x}{1-x^2}$$
- $$-\int P(x)\,dx = -\ln(1-x^2) \;\Rightarrow\; e^{-\int P\,dx} = \frac{1}{1-x^2}$$
$$v(x) = \int \frac{1}{x^2(1-x^2)}\,dx = \int \left(\frac{1}{x^2} + \frac{1}{1-x^2}\right)\,dx$$
$$v(x) = -\frac{1}{x} + \frac{1}{2}\ln\left|\frac{1+x}{1-x}\right|$$
$$\boxed{y_2 = -1 + \frac{x}{2}\ln\left|\frac{1+x}{1-x}\right|}$$



- **Problem 4.** Given $y_1 = \cos x$ solves $y'' + y = 0$, find y_2 by reduction of order and write the fundamental set for the solution space.
- Here $P(x) = 0$, so $e^{-\int P\,dx} = 1$.

$$v(x) = \int \frac{1}{\cos^2 x}\,dx = \int \sec^2 x\,dx = \tan x$$

$$y_2 = v(x)\,y_1(x) = \tan x \cdot \cos x = \sin x$$

$$\boxed{y_2 = \sin x}$$

- Thus, the fundamental set for the solution space of $y'' + y = 0$ is $\{\cos x, \sin x\}$.



Linear 2nd order DEs with Constant Coefficient

- Consider $y'' + ay' + by = 0,\ a, b \in \mathbb{R}$.
- Try a solution of the form $y = e^{rx}$.
- Substituting: $(r^2 + ar + b)e^{rx} = 0 \ \Rightarrow \ r^2 + ar + b = 0$.
- This is called the **characteristic (auxiliary) equation**.
- Its roots r_1, r_2 determine the form of the fundamental set of solutions.
- Three cases arise, based on the discriminant $a^2 - 4b$.
- Each case yields two linearly independent solutions.



2nd order LDE with CC

- **Case 1: Distinct Real Roots** $(a^2 - 4b > 0)$.
- Roots $r_1 \neq r_2$, both real.
- Fundamental set: $y_1 = e^{r_1 x}$, $y_2 = e^{r_2 x}$.
- Wronskian: $W(y_1, y_2)(x) = (r_2 - r_1)e^{(r_1+r_2)x} \neq 0$.
- Hence y_1, y_2 are linearly independent.
- General solution: $y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$.
- This covers overdamped-type behaviour in applications.



2nd order LDE with CC

- **Case 2: Repeated Real Root** ($a^2 - 4b = 0$).
- Root $r = r_1 = r_2 = -a/2$ (repeated).
- One solution: $y_1 = e^{rx}$.
- A second, independent solution is found by **reduction of order**: $y_2 = xe^{rx}$.
- Wronskian: $W(y_1, y_2)(x) = e^{2rx} \neq 0$, confirming independence.
- General solution: $y = c_1 e^{rx} + c_2 x e^{rx}$.
- This is the critically damped case in applications.



- **Case 3: Complex Conjugate Roots** ($a^2 - 4b < 0$).
- Roots $r_{1,2} = \alpha \pm i\beta$, with $\beta \neq 0$.
- Complex solutions $e^{(\alpha+i\beta)x}$, $e^{(\alpha-i\beta)x}$ combine via Euler's formula.
- Real fundamental set: $y_1 = e^{\alpha x}\cos(\beta x)$, $y_2 = e^{\alpha x}\sin(\beta x)$.
- Wronskian: $W(y_1, y_2)(x) = \beta e^{2\alpha x} \neq 0$ since $\beta \neq 0$.
- General solution: $y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$.
- This is the underdamped/oscillatory case in applications.



Examples

- **Q** Solve $y'' + 4y = 0$.
- Characteristic equation: $r^2 + 4 = 0 \Rightarrow r = \pm 2i$
- Fundamental set: $y_1 = \cos 2x,\ y_2 = \sin 2x$
- $W(y_1, y_2) = 2\cos^2 2x + 2\sin^2 2x = 2 \neq 0$

$$\boxed{y = c_1 \cos 2x + c_2 \sin 2x}$$

- **Q** Solve the IVP $y'' - 5y' + 6y = 0,\ y(0) = 1, y'(0) = 0$.
- Characteristic equation: $r^2 - 5r + 6 = 0 \Rightarrow r = 2, 3$
- General solution: $y = c_1 e^{2x} + c_2 e^{3x}$
- From $y(0) = 1$: $c_1 + c_2 = 1$
- From $y'(0) = 0$: $2c_1 + 3c_2 = 0 \Rightarrow c_1 = 3, c_2 = -2$

$$\boxed{y = 3e^{2x} - 2e^{3x}}$$



Example

- **Q** Solve $y'' - 6y' + 9y = 0$ subject to $y(0) = 2, \quad y'(0) = 5.$
- The auxiliary equation : $r^2 - 6r + 9 = 0 \Rightarrow r = 3, 3.$
- General solution: $y(x) = (C_1 + C_2 x)e^{3x}.$
- $y'(x) = e^{3x}\left(3C_1 + C_2 + 3C_2 x\right).$
- Using initial condition, $y(0) = 2$, we have $C_1 = 2.$
- Using initial condition, $y'(0) = 5$, we obtain $3C_1 + C_2 = 5.$
- Thus, $C_2 = 5 - 6 = -1.$
- Hence, the unique solution of the initial value problem is $y_p(x) = (2 - x)e^{3x}.$



Undetermined Coefficients - To compute y_p

- **Goal.** Find a particular solution y_p of $y'' + Py' + Qy = R(x)$ when P, Q are constants.
- Applicable only when $R(x)$ is a polynomial, exponential, sine/cosine, or a product/sum of these.
- **Idea:** Guess a trial form for y_p with unknown coefficients, matching the type of $R(x)$.
- Substitute the trial form into the ODE and compare coefficients to solve for the unknowns.
- Simpler than variation of parameters, but restricted to these special forcing terms.
- If $R(x) = R_1(x) + R_2(x)$, solve separately for each and add (**superposition**).
- Modification is required if the trial form *duplicates* a term already in y_c.



UC: Table of standard trial forms for y_p

- Assume that the trail form y_p has **no duplication with** y_c.
- $R(x) = P_n(x)$ (degree n polynomial)
  $\Rightarrow\ y_p = A_n x^n + \cdots + A_1 x + A_0$ **(0 is not root of AE).**
- $R(x) = ae^{kx} \ \Rightarrow\ y_p = Ae^{kx}$ **(k is not root of AE).**
- $R(x) = a\cos\beta x$ or $a\sin\beta x \ \Rightarrow\ y_p = A\cos\beta x + B\sin\beta x$
  ($i\beta$ is not root of AE).
- $R(x) = P_n(x)e^{kx} \ \Rightarrow\ y_p = (A_n x^n + \cdots + A_0)e^{kx}$
  (k is not root of AE).
- $R(x) = e^{kx}(a\cos\beta x + b\sin\beta x) \ \Rightarrow\ y_p =$
  $e^{kx}(A\cos\beta x + B\sin\beta x)$ **(k+$i\beta$ is not root of AE).**
- $R(x) = P_n(x)e^{kx}\cos\beta x$ (or sin)
  $\Rightarrow\ y_p = e^{kx}\left[(A_n x^n + \cdots)\cos\beta x + (B_n x^n + \cdots)\sin\beta x\right]$
  (k+$i\beta$ is not root of AE).



UC: Modification in the trial form for y_p

- Let s be the multiplicity of the exponent k (or $k \pm i\beta$) as a root of the characteristic equation.
- If $s = 0$: no overlap; use the standard trial form as described in the previous slide.
- If $s = 1$: the standard trial form solves the homogeneous equation, so multiply the whole trial by x. Then, $y_p = xy_t$, where y_t is assumed based on the previous slide.
- If $s = 2$: (repeated root case) multiply the whole trial by x^2. Then, $y_p = x^2 y_t$, where y_t is assumed based on the previous slide.
- In general, multiply by x^s, where s is the number of times the root is repeated.
- This ensures y_p is *not* itself a solution of the homogeneous equation (not part of y_c).



Step-by-Step Procedure for the use of UC

1. Consider the associated homogeneous system.
2. Compute the Auxiliary Equation.
3. Solve the homogeneous equation to find y_c.
4. Choose the trial form for y_p based on $R(x)$'s type (see table).
5. Check for duplication with y_c; multiply by x^s if needed.
6. Substitute y_p into the ODE; equate coefficients of like terms.
7. Solve the resulting linear system for the unknown coefficients.
8. Write the general solution $y = y_c + y_p$.



Example

- Solve $y'' - 3y' + 2y = e^{3x}$.
- Roots: $r = 1, 2$; $y_c = c_1 e^x + c_2 e^{2x}$.
- Since 3 is not a root of AE, $s = 0$: trial $y_p = Ae^{3x}$.
- Now $y_p = Ae^{3x}$ satisfies the given non-homogeneous DE.
- Substituting: $9A - 9A + 2A = 2A$, so $2A = 1 \Rightarrow A = \frac{1}{2}$.

The PS is $y_p = \dfrac{1}{2} e^{3x}$

The GS is $\boxed{y = y_c + y_p = c_1 e^x + c_2 e^{2x} + \dfrac{1}{2} e^{3x}.}$



Examples

- Solve $y'' - 3y' + 2y = e^{2x}$.
- Roots: $r = 1, 2$; since 2 *is* a simple root and $s = 1$, we take $y_p = Axe^{2x}$. (Note $y_t = Ae^{2x}$).
- Substituting and simplifying: $y_p'' - 3y_p' + 2y_p = Ae^{2x}$.
- Setting $Ae^{2x} = e^{2x}$ gives $A = 1$.
  The PS is $y_p = xe^{2x}$
  GS is $\boxed{y = c_1 e^x + c_2 e^{2x} + xe^{2x}}$.
- Solve $y'' - 4y' + 4y = e^{2x}$.
- Roots: $r = 2, 2$ (repeated). Since $s = 2$, we take $y_p = Ax^2 e^{2x}$ (Note $y_t = Ae^{2x}$).
- Substituting and simplifying: $y_p'' - 4y_p' + 4y_p = 2Ae^{2x}$.
- Setting $2Ae^{2x} = e^{2x}$ gives $A = \frac{1}{2}$.
  The PS is $y_p = \dfrac{1}{2}x^2 e^{2x}$.
  GS is $\boxed{y = (c_1 + c_2 x)e^{2x} + \dfrac{1}{2}x^2 e^{2x}.}$

